```
======================================================================

DATASET 2: Daily Stock Returns

======================================================================

Date range: 2023-01-01 to 2024-05-14

Number of days: 500



Price Statistics:

count    500.000000

mean      98.201636

std       10.934234

min       80.570476

25%       87.977257

50%       97.999752

75%      106.893544

max      127.558311

Name: Price, dtype: float64



----------------------------------------------------------------------

B1: Stationarity Tests

----------------------------------------------------------------------

Saved: B1_stationarity.png

Visual inspection: Rolling mean changes over time → NOT stationary



ADF Test on Prices:

  Test Statistic = -1.4402
```

```
  p-value        = 0.5628

  Lags used      = 0

  Decision (α=0.05): Fail to reject H0 → Non-stationary



KPSS Test on Prices:

  Test Statistic = 2.8730

  p-value        = 0.0100

  Decision (α=0.05): Reject H0 → Non-stationary



Interpretation:

  ADF: Fails to reject → suggests non-stationary (has unit root)

  KPSS: Rejects H0 → confirms non-stationary

  Both tests agree: the PRICE series is NON-STATIONARY.



----------------------------------------------------------------------

B2: Differencing for Stationarity

----------------------------------------------------------------------

Saved: B2_1_differenced.png

ADF Test on Differenced Prices:

  Test Statistic = -20.8671

  p-value        = 0.000000

  Decision: Reject H0 → Stationary



Correlation between differenced prices and returns: 0.993756

Differenced prices ≈ returns × price_level (approximately proportional)

Saved: B2_2_acf_diff.png
```

```
Most ACF lags within confidence bands → differenced series is approximately white noise.



----------------------------------------------------------------------

B3: ACF and PACF

----------------------------------------------------------------------

Saved: B3_acf_pacf.png

ACF Interpretation:

  - Most lags within confidence bands

  - Returns show very weak autocorrelation (near white noise)

  - ACF does not show clear exponential decay or sharp cutoff



PACF Interpretation:

  - Similar pattern; most lags insignificant

  - Possible small spike at lag 1



Suggested orders: AR(0) or AR(1) for p; MA(0) or MA(1) for q



ρ(1) = Corr(r_t, r_{t-1}) = 0.0455

ρ(5) = Corr(r_t, r_{t-5}) = 0.0302



----------------------------------------------------------------------

B4: AR(p) Model

----------------------------------------------------------------------

=== AR(1) Summary ===

                               SARIMAX Results

==============================================================================
```

```
Dep. Variable:                Returns   No. Observations:                  500
Model:                  ARIMA(1, 0, 0)   Log Likelihood                -902.165
Date:                 Tue, 10 Feb 2026   AIC                           1810.330
Time:                         21:28:30   BIC                           1822.973
Sample:                     01-01-2023   HQIC                          1815.291
                          - 05-14-2024
Covariance Type:                   opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0104      0.069      0.149      0.882      -0.126       0.147
ar.L1          0.0455      0.047      0.976      0.329      -0.046       0.137
sigma2         2.1614      0.130     16.678      0.000       1.907       2.415
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 4.13
Prob(Q):                              0.99   Prob(JB):                         0.13
Heteroskedasticity (H):               1.17   Skew:                             0.18
Prob(H) (two-sided):                  0.31   Kurtosis:                         3.26
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

ϕ_1 = 0.0455

Significant? p-value = 0.3290 → No

Sign: Positive (momentum)

```
|ϕ₁| = 0.0455 < 1 → Stationary


--- AR Model Comparison ---

Model              AIC        BIC

------------------------------

AR(1)          1810.33    1822.97

AR(2)          1812.30    1829.15

AR(3)          1814.25    1835.32

Saved: B4_ar1_residuals.png

Residuals appear to be white noise (no significant ACF spikes).


----------------------------------------------------------------------

B5: MA(q) Model

----------------------------------------------------------------------

=== MA(1) Summary ===

                               SARIMAX Results

==============================================================================

Dep. Variable:                Returns   No. Observations:                  500

Model:                 ARIMA(0, 0, 1)   Log Likelihood                -902.157

Date:                Tue, 10 Feb 2026   AIC                           1810.313

Time:                        21:28:30   BIC                           1822.957

Sample:                    01-01-2023   HQIC                          1815.275

                         - 05-14-2024

Covariance Type:                  opg

==============================================================================

                 coef    std err          z      P>|z|      [0.025      0.975]
```

```
------------------------------------------------------------------------------

const          0.0104      0.069      0.149      0.881     -0.126      0.146

ma.L1          0.0463      0.047      0.993      0.321     -0.045      0.138

sigma2         2.1614      0.130     16.682      0.000      1.907      2.415

==================================================================================

Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 4.16

Prob(Q):                              1.00   Prob(JB):                         0.12

Heteroskedasticity (H):               1.17   Skew:                             0.18

Prob(H) (two-sided):                  0.31   Kurtosis:                         3.26

==================================================================================


Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).


θ₁ = 0.0463

Significant? p-value = 0.3208

|θ₁| = 0.0463 < 1 → Invertible


--- AR(1) vs MA(1) ---

Metric                 AR(1)        MA(1)

----------------------------------------

AIC                  1810.33      1810.31

BIC                  1822.97      1822.96

Log-Lik              -902.16      -902.16


Better model by AIC: MA(1)
```

```
Saved: B5_ma1_residuals.png



----------------------------------------------------------------------

B6: ARMA(p,q) Model

----------------------------------------------------------------------

=== ARMA(1,1) Summary ===

                               SARIMAX Results

==============================================================================

Dep. Variable:                Returns   No. Observations:                  500

Model:                 ARIMA(1, 0, 1)   Log Likelihood                -901.516

Date:                Tue, 10 Feb 2026   AIC                           1811.031

Time:                        21:28:30   BIC                           1827.889

Sample:                    01-01-2023   HQIC                          1817.646

                         - 05-14-2024

Covariance Type:                  opg

==============================================================================

                 coef    std err          z      P>|z|      [0.025      0.975]

------------------------------------------------------------------------------

const          0.0105      0.068      0.154      0.878      -0.123       0.144

ar.L1         -0.7754      0.259     -2.995      0.003      -1.283      -0.268

ma.L1          0.8186      0.237      3.451      0.001       0.354       1.283

sigma2         2.1558      0.129     16.665      0.000       1.902       2.409

===================================================================================

Ljung-Box (L1) (Q):                   0.02   Jarque-Bera (JB):                 5.23

Prob(Q):                              0.89   Prob(JB):                         0.07

Heteroskedasticity (H):               1.20   Skew:                             0.20
```

```
Prob(H) (two-sided):                  0.25   Kurtosis:                       3.30

==============================================================================



Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



--- Overparameterization Check ---

ϕ₁ p-value: 0.0027 → Significant

θ₁ p-value: 0.0006 → Significant



--- Full Model Comparison ---

Model                AIC        BIC    Log-Lik   Params

-----------------------------------------------------

AR(1)            1810.33    1822.97    -902.16        3

MA(1)            1810.31    1822.96    -902.16        3

ARMA(1,1)        1811.03    1827.89    -901.52        4



----------------------------------------------------------------------

B7: Automatic ARIMA

----------------------------------------------------------------------

Top 10 models by AIC:

 p  q         AIC         BIC

 2  4 1807.845833 1841.562697

 4  2 1808.061214 1841.778079

 0  0 1809.364768 1817.793985

 3  4 1809.853316 1847.784789
```

```
 2  5 1810.189389 1848.120862

 0  1 1810.313437 1822.957261

 1  0 1810.329607 1822.973431

 1  1 1811.031039 1827.889472

 3  2 1812.118870 1841.621127

 0  2 1812.281349 1829.139782


Best model: ARIMA(2, 0, 4) with AIC = 1807.85


=== Best Model ARIMA(2, 0, 4) Summary ===

                               SARIMAX Results

==============================================================================

Dep. Variable:                Returns   No. Observations:                  500

Model:                 ARIMA(2, 0, 4)   Log Likelihood                -895.923

Date:                Tue, 10 Feb 2026   AIC                           1807.846

Time:                        21:28:36   BIC                           1841.563

Sample:                    01-01-2023   HQIC                          1821.076

                         - 05-14-2024

Covariance Type:                  opg

==============================================================================

                 coef    std err          z      P>|z|      [0.025      0.975]

------------------------------------------------------------------------------

const          0.0133      0.068      0.196      0.845      -0.120       0.147

ar.L1          0.0198      0.022      0.886      0.376      -0.024       0.063

ar.L2         -0.9294      0.023    -40.883      0.000      -0.974      -0.885

ma.L1          0.0245      0.052      0.469      0.639      -0.078       0.127
```

```
ma.L2          0.9474      0.092     10.309      0.000       0.767       1.128

ma.L3          0.0565      0.049      1.143      0.253      -0.040       0.153

ma.L4         -0.0495      0.051     -0.981      0.327      -0.149       0.049

sigma2         2.0910      0.198     10.541      0.000       1.702       2.480

==================================================================================

Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 4.29

Prob(Q):                              0.98   Prob(JB):                         0.12

Heteroskedasticity (H):               1.14   Skew:                             0.14

Prob(H) (two-sided):                  0.41   Kurtosis:                         3.36

==================================================================================



Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).



Does auto-selection match manual? The best model is ARIMA(2, 0, 4).

Given the weak autocorrelation in returns, a simple model (like AR(1) or even

ARIMA(0,0,0)) is expected to perform similarly.



----------------------------------------------------------------------

B8: ARIMA Forecasting

----------------------------------------------------------------------

Saved: B8_1_return_forecast.png

Saved: B8_2_price_forecast.png



Forecast Behavior Observations:

  - Return forecasts converge to the unconditional mean (-0.0575%)
```

```
    as horizon increases, since AR effects decay exponentially.

  - Prediction interval width at h=1:  5.6754

  - Prediction interval width at h=20: 5.7448

  - Intervals widen because forecast uncertainty accumulates with horizon.

  - For near-white-noise returns, intervals widen quickly and forecasts

    offer little improvement over the unconditional mean.



======================================================================

PART B COMPLETE - All 8 plots saved!

======================================================================
```